

INTERVIEW GUIDE

50 Key Terms for AI Interviews

Essential concepts you need to know for Agentic AI, Generative AI, and RAG interviews

Agentic AI

RAG

LLMs

Generative AI

Frameworks

Interview Preparation



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1. Agentic AI

AI systems that operate autonomously to achieve goals without human guidance.

2. ReAct Pattern

Agent architecture that interleaves reasoning steps with action execution.

3. Tool Use

LLM capability to interact with external systems like APIs and databases.

4. Function Calling

Structured method for LLMs to invoke functions with defined schemas.

5. Memory Management

System for agents to maintain context and learn from interactions.



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6. Short-term Memory

Working memory that holds current conversation context within token limits.

7. Long-term Memory

Persistent storage using vector databases to remember information across sessions.

8. Planning Strategies

Methods for agents to decompose complex goals into achievable step plans.

9. Chain-of-Thought

Prompting technique that makes LLMs reason step-by-step before answering.

10. Tree-of-Thoughts

Planning approach that explores multiple reasoning paths and evaluates branches.



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11. Plan-and-Execute

Pattern that separates planning from execution, creating complete plans first.

12. Error Recovery

Agent ability to gracefully handle failures and retry with backoff.

13. Fault Tolerance

System resilience to handle failures through fallback mechanisms and checkpoints.

14. Multi-Agent Systems

Multiple specialized agents working together to solve complex tasks.

15. Agent Orchestration

Coordination of multiple agents with defined workflows and communication protocols.



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16. State Management

Tracking what the agent remembers and maintains across interactions.

17. Human-in-the-Loop

Design pattern requiring human approval for sensitive actions before execution.

18. Prompt Injection

Security attack where malicious inputs override agent instructions and behavior.

19. Output Validation

Verification that tool results and LLM outputs are sensible before proceeding.

20. Checkpointing

Saving agent state at critical points to enable recovery from failures.



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21. LangChain

Python framework for building LLM applications with chains and agents.

22. LangGraph

Framework for building stateful multi-agent applications with graph workflows.

23. CrewAI

Python framework for orchestrating multiple AI agents working as a crew.

24. Autogen

Microsoft framework for building multi-agent conversational AI systems.

25. Agent Performance Evaluation

Measuring task success rate, efficiency, quality, and safety metrics.



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26. RAG

Retrieval Augmented Generation technique that enhances LLMs with external knowledge.

27. Embeddings

Numerical vector representations of text that capture semantic meaning.

28. Vector Databases

Specialized databases optimized for storing and querying high-dimensional embeddings.

29. Chunking

Splitting documents into smaller segments for indexing and retrieval in RAG.

30. Hybrid Search

Combining dense vector retrieval with sparse keyword matching for better results.



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31. Reranking

Re-scoring retrieved documents using cross-encoders to improve relevance.

32. Dense Retrieval

Semantic search using embeddings to find conceptually similar documents.

33. Sparse Retrieval

Keyword-based search using algorithms like BM25 for exact term matching.

34. Semantic Search

Finding documents based on meaning rather than exact keyword matches.

35. Context Compression

Reducing retrieved context size while preserving important information.



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36. Transformers

Neural architecture using self-attention to process sequences in parallel.

37. Self-Attention

Mechanism allowing each position to attend to all other positions.

38. Large Language Models

Massive neural networks trained on vast text data to generate text.

39. Fine-tuning

Training pre-trained models on domain-specific data to improve performance.

40. Prompt Engineering

Crafting effective prompts to guide LLM behavior and improve output quality.



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41. Hallucination

LLM generating false or fabricated information that seems plausible.

42. Token Limits

Maximum number of tokens an LLM can process in a single request.

43. Context Window

Maximum sequence length an LLM can consider when generating responses.

44. Few-shot Learning

LLM ability to learn from a small number of examples in prompt.

45. Zero-shot Learning

LLM ability to perform tasks without any training examples.



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46. Diffusion Models

Generative models that create data by reversing a gradual noising process.

47. GANs

Generative Adversarial Networks with generator and discriminator competing in training.

48. Latent Space

Lower-dimensional representation space where generative models encode meaningful features.

49. Mode Collapse

GAN problem where generator produces limited variety focusing on few patterns.

50. Evaluation Metrics

Quantitative measures like accuracy and F1 score for assessing AI systems.



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